Tanmay Bhuskute

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PROFESSIONAL SUMMARY

Software Engineer with experience designing, automating, and operating production cloud infrastructure at Siemens. Built deployment platforms, Kubernetes workflows, and CI/CD automation that improved deployment speed and reliability. Currently researching LLMs for Android testing and bug reproduction as a Master's student at the University of Cincinnati.

EDUCATION

Master of Science in Computer Science - University of Cincinnati — Aug 2024 - Dec 2026
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering - AISSMS College of Engineering — Aug 2018 - Jul 2022
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming & Scripting: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, Go, Rust.
Frameworks: React, Node.js, JUnit, FastAPI, Streamlit, Django, Flask, BERT.
Developer Tools: VS Code, Git, Docker, AWS, Azure, Google Cloud Platform, SonarQube, Postman, JFrog, Polarion, Jenkins, ArgoCD, Kubernetes, REST APIs, Linux, Unix, Bash, TensorFlow.
Libraries: PyTorch, scikit-learn, pandas, NumPy, matplotlib, HuggingFace, OpenAI, LangChain, Pydantic, spaCy.

WORK EXPERIENCE

Siemens Industry Software (Software Developer Engineer) — Jul 2022 - Jun 2024
Tech Stack: AWS, Python, Boto3, Kubernetes, ArgoCD, Helm, HashiCorp Vault

- Executed **44 production environment upgrades in 4 months** across Dockerized, **Kubernetes**-based Teamcenter X services, ensuring reliable deployments through careful planning, validation, and rollback procedures.
- Migrated **40+ production environments** from process-based deployments to GitOps-driven **Kubernetes** workflows using **ArgoCD**, **Helm**, Amazon EFS, and HashiCorp Vault, modernizing deployment architecture.
- Designed and implemented an Ansible-based Blue-Green deployment framework with automated rollback, **reducing production upgrade downtime by 50%** and enabling full environment recovery within **15 minutes**.
- Automated **AWS infrastructure workflows** using **Python (Boto3)**, replacing legacy Ruby scripts and **reducing deployment time by 40%** across production environments.
- Reduced monthly **AWS infrastructure costs by 40% (from $15K to $9K)** through EC2 rightsizing and automated cleanup of unused cloud resources using CloudWatch utilization metrics.
- Built a self-service developer platform (dashboard, CLI, and YAML automation) for environment lifecycle management, **adopted by 15 engineers**, **reducing environment provisioning time by 70%** and lifecycle operations by up to **90%**.
- Served as Quality Coach and SonarQube Specialist for the Agile Release Train, **increasing automated test coverage from 35% to 78%** by enforcing coding standards and CI/CD quality gates.

Siemens Industry Software (Software Developer Intern) — Mar 2022 - Jun 2022
Tech Stack: AWS, Terraform, Ansible, Terratest, Go

- Automated AWS infrastructure provisioning with Terraform and optimized Ansible-based DevOps workflows, **reducing deployment time by 30%** and environment setup from hours to minutes.
- Developed a Go + Terratest validation framework that **caught 95% of infrastructure misconfigurations** before production, improving deployment reliability.

Government of Maharashtra, Water Resource Department (Software Developer Intern) — Sep 2021 - Dec 2021
Tech Stack: React.js, MySQL, Full-Stack Development, UI/UX Design, Process Automation

- Developed full-stack **React/MySQL** register app, digitizing record management and automating workflows.
- **Improved efficiency by 40%**, deploying a system that improved transparency and accuracy across the department.

PROJECTS AND RESEARCH

Memory-Augmented LLM Approach for Android Bug Reproduction from Video Reports — Feb 2026 - Present
Tech Stack: Python, Google Gemini 2.5 Pro, Vertex AI, ADB, uiautomator2

- Proposed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android bug reproduction.
- Demonstrated **25.7x latency reduction** vs baseline ViBR across **20+ Android apps**.

PolicyPilot - Medical Benefit Policy Intelligence Portal (Agents Assemble Hackathon) — April 2026
Tech Stack: React, TypeScript, Node.js, Google Gemini, MCP, Docker

- Architected evidence-grounded **RAG** pipeline over **Google Gemini 2.5-flash**, enforcing citation-backed answers to eliminate hallucination in eligibility decisions.
- Designed **17-tool MCP** server for PA extraction, readiness evaluation, and payer comparison, exposed to external AI agents.

EXTRACURRICULAR ACTIVITIES

- Finalist at RevolutionUC 2025 Hackathon, among **300 participants from 31 schools**.
- Earned **SAFe Practitioner 5.1 Certification** (Mar 2023).